

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2021
CHE-1104: Chemistry

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Describe Bohr's atom model. Mention its limitations. 4
 b) What are quantum numbers? Explain the physical significance of the four quantum numbers. 4
 c) Discuss the following rules for electronic configuration of atom: 4
 (i) Pauli's exclusion principle (ii) Hund's rule
 d) Write the electronic configuration of the following atoms or ions 2
 Sc, Cr, Fe^{2+} , Br^-
2. a) Derive the ideal gas equation and state its limitations. 5
 b) Explain- 'Electron affinity of Fluorine is less than that of Chlorine.' 3
 c) Determine the electronic configuration of the followings- 3
 Fe^{3+} , Cu, O^{2-}
 d) Why the aqueous solution of $\text{Al}_2(\text{SO}_4)_3$ is acidic in nature? 3
3. a) What is chemical bonding? Describe the electronic theory of chemical bond. 4
 b) Explain why CH_4 , NH_3 and H_2O molecules have different shapes? 4
 c) Explain the followings- 6
 (i) NCl_3 does not exist while PCl_3 exists.
 (ii) H_2O is a liquid but H_2S is a gas at ordinary temperature.
4. a) Declare the rate expression for the type of the following reaction: 4
 $2\text{A} \longrightarrow \text{Products}$.
 b) Why a first order reaction will never be completed? 4
 c) What is buffer solution? Write the mechanism of acid buffer action. 4
 d) What is activation energy? 2
5. a) Explain the dynamic character of chemical equilibrium. 3
 b) If α is the amount of degree of dissociation then express the K_p of the following reaction. 4
 $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
 c) At 500°C , the reaction between N_2 and H_2 to form ammonia has $K_c = 6.0 \times 10^{-2}$. What is the numerical value of K_p for the reaction? 3
 d) Explain with the help of Le-Chatelier's principle the effect of temperature and pressure on the reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{heat}$. 4
6. a) What is hybridization? Discuss the hybridization of Methane. 3
 b) Discuss the enthalpy of a reaction and state Hess's law of thermochemistry. 5
 c) Calculate ΔH° for the reaction: $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g})$ 4
 Given that of ΔH° for $\text{CO}_2(\text{g})$, $\text{CO}(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ are -393.5 , -111.31 and -241.80 kJ mol^{-1} respectively.
 d) Determine oxidizing and reducing agents from the following reaction- 2
 $\text{Zn}(\text{s}) + \text{CuSO}_4(\text{aq}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu}(\text{s})$
7. a) Define standard electrode potential and state the difference between Electrochemical cell and Electrolytic cell. 3
 b) Briefly explain the mechanism of a voltaic cell. 6
 c) Calculate the emf of the cell: $\text{Zn} | \text{Zn}^{2+}(0.001\text{M}) || \text{Ag}^+(0.1\text{M}) | \text{Ag}$. 3
 Given the standard potential of Ag^+/Ag half-cell is $+0.80$ V and Zn^{2+}/Zn is -0.76 V.
 d) What is Gibb's phase rule? 2